

An exact count for OEIS A228462: windowed maxima over a growing alphabet

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Abstract

OEIS A228462 counts the arrays obtainable as the maxima of 3 adjacent elements of a length 7 array with entries in $\{0, \dots, n\}$, and carries an empirical polynomial for that count. The count is not a transfer-matrix count: the length is fixed and it is the ALPHABET that grows with n . It is nevertheless exactly computable, because membership in the image depends only on the relative order of the entries of the candidate array and not on their values, so the count is a fixed non-negative combination of binomial coefficients $\binom{n+1}{m}$. Computing that combination settles the empirical claim as an identity in $\mathbb{Q}[n]$.

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1 The conjecture

The entry's first terms are 17, 91, 310, 821, 1847, 3703, 6812, 11721, ... and it states

$$\text{Empirical: } a(n) = (2/15)n^5 + (7/6)n^4 + (25/6)n^3 + (19/3)n^2 + (21/5)n + 1.$$

2 The object

Fix $k = 7$ and $w = 3$, and put $L = k - w + 1 = 5$. For $a = (a_1, \dots, a_k) \in \{0, \dots, n\}^k$ let

$$\Phi(a) = (\max(a_1, \dots, a_w), \max(a_2, \dots, a_{w+1}), \dots, \max(a_{k-w+1}, \dots, a_k)) \in \{0, \dots, n\}^L.$$

The entry counts $|\Phi(\{0, \dots, n\}^k)|$.

3 Membership in the image is decided by the order type

Write $W_j = \{j, j+1, \dots, j+w-1\}$ for the j -th window, $1 \leq j \leq L$.

Lemma 1. *Let $b \in \{0, \dots, n\}^L$ and define $\hat{a} \in \{0, \dots, n\}^k$ by $\hat{a}_i = \min\{b_j : i \in W_j\}$. Then $b \in \Phi(\{0, \dots, n\}^k)$ if and only if $\Phi(\hat{a}) = b$.*

Proof. Each $\hat{a}_i$ is one of the b_j , so $\hat{a}$ does lie in $\{0, \dots, n\}^k$ and the condition is meaningful. If $\Phi(\hat{a}) = b$ there is nothing to prove. Conversely suppose $\Phi(a) = b$ for some a . For every $i \in W_j$ we have $a_i \leq \max_{l \in W_j} a_l = b_j$, so $a_i \leq \min\{b_j : i \in W_j\} = \hat{a}_i$: the array $\hat{a}$ dominates every witness componentwise. Hence $\max_{i \in W_j} \hat{a}_i \geq \max_{i \in W_j} a_i = b_j$. The reverse inequality is immediate from the definition, since $\hat{a}_i \leq b_j$ for each $i \in W_j$. So $\Phi(\hat{a}) = b$. $\square$

Both the construction of $\hat{a}$ and the test $\Phi(\hat{a}) = b$ use only comparisons between entries of b . Consequently:

Corollary 2. *Whether b lies in the image depends only on the order type of b – the weak ordering of its L coordinates – and not on n or on the values themselves.*

Proof. Let σ be any strictly increasing map defined on the values of b . Applying σ entrywise commutes with $\min$ and $\max$, so it carries $\hat{a}$ for b to $\hat{a}$ for $\sigma(b)$ and preserves the equalities tested in Lemma 1. Any two arrays with the same order type differ by such a σ . $\square$

4 The count

For $1 \leq m \leq L$ let A_m be the number of order types of L coordinates that use exactly m distinct values and are achievable. An array $b \in \{0, \dots, n\}^L$ using exactly m distinct values is determined by the m -element subset of $\{0, \dots, n\}$ it uses together with its order type, and every pair occurs exactly once; so by Corollary 2,

$$|\Phi(\{0, \dots, n\}^k)| = \sum_{m=1}^5 A_m \binom{n+1}{m} \quad \text{for every } n \geq 0.$$

Enumerating the weak orderings of 5 elements and applying Lemma 1 to each gives $A_1 = 1$, $A_2 = 15$, $A_3 = 43$, $A_4 = 44$, $A_5 = 16$, so

$$|\Phi(\{0, \dots, n\}^k)| = 1 \binom{n+1}{1} + 15 \binom{n+1}{2} + 43 \binom{n+1}{3} + 44 \binom{n+1}{4} + 16 \binom{n+1}{5} = \frac{2n^5}{15} + \frac{7n^4}{6} + \frac{25n^3}{6} + \frac{19n^2}{3} + \frac{2n}{3} + \frac{1}{3}.$$

5 Conclusion

The right-hand side is the polynomial the entry states, so the empirical claim of OEIS A228462 is true for every $n \geq 0$. The argument is finite and exact: it enumerates the weak orderings of 5 elements, of which there are 119 achievable, and performs no fitting to the published terms. Those terms are reproduced by the formula as a check, not as evidence.